Replicating xRFM: Accurate, Scalable, and Interpretable Feature Learning Models for Tabular Data +bonus

COMP9417 Group Project

Xueyao Jiang z5581908 Chang Yang z5569348 Binyue Peng z5705242
Jiayu Zhang z5649264 Zirun Zhang z5617187

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1 Introduction

Tabular data remains one of the most common data modalities in applied machine learning, appearing in domains such as materials science, healthcare, e-commerce, and human-resource analytics. Despite the rapid progress of deep learning on images and language, GBDTs such as XGBoost and LightGBM remain difficult to outperform on many tabular tasks [Grinsztajn et al., 2022, Holzmüller et al., 2024].

This project replicates and evaluates xRFM, a recent feature learning approach for tabular data based on the average gradient outer product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. Instead of relying only on fixed input features, xRFM uses gradient-based information from a learned prediction function to estimate directions in the input space that are most relevant to the target. This produces both a predictive model and an interpretable feature-relevance signal. The central question of our replication is therefore whether AGOP-driven feature learning can offer a competitive alternative to strong tabular baselines, especially when compared with XGBoost, LightGBM, and a multilayer perceptron (MLP).

We evaluate all models under a unified pipeline across five public tabular datasets covering regression and binary classification, large and small sample sizes, and mixed feature types (Diamonds, Superconductivity, Online Shoppers, Stroke, HR Attrition). Our main finding is that tree-based GBDTs lead overall, while xRFM shows selective strength on small-sample and fully numerical tasks. Section 2 describes methodology, Section 3 reports results, and Sections 4–5 discuss findings and conclusions.

2 Methodology

2.1 xRFM: Algorithm and Theoretical Background

xRFM is a tabular feature-learning model based on the Average Gradient Outer Product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. For a trained prediction function $f : \mathbb{R}^d \to \mathbb{R}$, the AGOP is defined as

$$G_f = \mathbb{E}_x\left[\nabla f(x) \nabla f(x)^\top\right] \in \mathbb{R}^{d \times d}. \tag{1}$$

The diagonal entry $[G_f]_{ii} = \mathbb{E}_x[(\partial f/\partial x_i)^2]$ is the expected squared partial derivative with respect to feature i: a large value indicates rapid change in the model output along coordinate i (important feature); a small value indicates near-insensitivity. The full G_f supports two interpretability strategies: its *diagonal* gives coordinate-wise importance scores for each feature, while its *top eigenvectors* identify multivariate directions in input space along which the model varies most, potentially capturing signal invisible to any single coordinate. We use diagonal AGOP in our experiments as it is cheaper to store and sufficient for feature-level rankings.

Compared with alternative methods, AGOP is supervised and model-specific: it reflects the gradient geometry of the fitted function, unlike PCA which only uses input covariance [Pearson, 1901]. Mutual information measures marginal feature–target dependence but not the model's multivariate sensitivity [Kraskov et al., 2004]. Permutation importance is model-agnostic but measures post-hoc performance degradation rather than learned geometry [Fisher et al., 2019]. AGOP is extracted as a by-product of training and directly summarises which directions the learned model relies on.

Training pipeline. xRFM builds a binary tree over the training data. At each node, a base model f is fitted to the local data partition and its AGOP G_f is computed. The leading eigenvector v_1 of G_f defines the split direction; the threshold t^* minimises weighted target

variance along the projection $v_1^\top x$. Each leaf fits a local random Fourier feature kernel machine with the feature metric reweighted by G_f. For inference on a new point x^*, the tree is traversed via $v_k^\top x^* \leq t_k$ at each internal node, and the leaf model produces the final prediction. The dominant training cost is AGOP computation: $O(nd)$ per node for a linear approximation, giving total complexity $O(nd \cdot 2^D)$ for tree depth D [Beaglehole et al., 2025]. This addresses the fundamental limitation of standard kernel machines on tabular data — that a global Euclidean metric treats all features equally — by adaptively reweighting feature space at each tree node.

2.2 Experimental Design

2.2.1 Datasets

Five public tabular datasets (Table 1) are used, spanning regression and binary classification, small and large sample sizes, and mixed feature types [Hamidieh, 2018, Sakar et al., 2019, fedesoriano, 2021].

Table 1: Dataset statistics. n_{train}, n_{val}, and n_{test} denote the sizes of the training, validation, and test splits respectively; d is the number of features after preprocessing.

Dataset	Task	Target	n_{train}	n_{val}	n_{test}	d
Diamonds	Regression	price	32,365	10,789	10,789	26
Superconductivity	Regression	critical_temp	12,757	4,253	4,253	81
Online Shoppers	Classification	Revenue	7,398	2,466	2,466	27
Stroke	Classification	stroke	3,066	1,022	1,022	21
HR Attrition	Classification	Attrition	882	294	294	51

2.2.2 Preprocessing and Splitting

All experiments use seed 42 and a 60/20/20 split. Numerical features are z-score normalised and categorical features are one-hot encoded; missing values are median/constant imputed. The same encoded matrix is used across all four models.

2.2.3 Evaluation Metrics

Regression tasks are evaluated with RMSE; binary classification tasks with accuracy and AUC-ROC. Training and inference time are also recorded.

2.2.4 xRFM Configuration

The xRFM experiments use the Python `xrfm` package through the shared project training pipeline. The model is initialised as `xRFM(use_diag=True, device=DEVICE)`, where `DEVICE` is CUDA if available and CPU otherwise. On GPU, the training script sets the number of fit iterations to 3 and `M_batch_size=128` to control memory cost. Feature importance is extracted by reading the learned AGOP matrix or diagonal values from the trained xRFM model and ranking features by their AGOP-derived scores.

Table 2: xRFM configuration used in the experiments.

Component	Setting
Model	`xRFM`
AGOP mode	Diagonal AGOP, `use_diag=True`
Device	CUDA if available; otherwise CPU
Random seed	42
GPU fit iterations	3
GPU AGOP batch size	128
Data split	60/20/20 train/validation/test
Regression metric	RMSE
Classification metrics	Accuracy and AUC-ROC
Interpretability output	Top features ranked by AGOP diagonal score

2.2.5 XGBoost and LightGBM Configuration

XGBoost and LightGBM are trained through the scikit-learn API using strong default settings from recent tabular benchmarking work [Chen and Guestrin, 2016, Ke et al., 2017, Holzmüller et al., 2024]. Both use 1,000 maximum boosting rounds, learning rate $\eta = 0.05$, row and column subsampling of 0.8, and early stopping with patience 20 on the validation set. XGBoost uses maximum tree depth 6, while LightGBM uses `num_leaves=31`.

2.2.6 MLP Configuration

The MLP model uses a two-hidden-layer network with hidden sizes $(128, 64)$ and ReLU activation function. The model is trained using the Adam optimizer with a learning rate of 0.001, a batch size of 256, a maximum of 500 iterations, and an early stopping mechanism with a patience of 20. Input features are standardised before training. For classification tasks, predicted probabilities are used to calculate the AUC-ROC.

3 Results

This section compares xRFM with XGBoost, LightGBM, and MLP in terms of predictive performance, computational efficiency, scalability, and interpretability. All models are evaluated using the same preprocessing pipeline, train-validation-test split, and evaluation metrics described in Section 2.

3.1 Predictive Performance

Table 3 reports the final held-out test performance across all five datasets. RMSE is used for regression tasks, while accuracy and AUC-ROC are used for binary classification tasks. Bold values indicate the best result for each metric.

Regression tasks. LightGBM achieves the lowest RMSE on both regression datasets. xRFM is competitive on Superconductivity, where its RMSE of 9.454 is close to LightGBM's 9.209 and XGBoost's 9.302. However, xRFM performs much worse on Diamonds, with RMSE 1496.72 compared with 535.99 for LightGBM and 544.68 for XGBoost. This suggests that AGOP-based feature learning does not automatically outperform tree ensembles on large mixed-type tabular regression.

Table 3: Test-set performance of all models. **Bold** indicates the best result per dataset and metric. RMSE is lower-better; Accuracy and AUC are higher-better.

Dataset	Metric	Model			
		xRFM	XGBoost	LightGBM	MLP
Diamonds	RMSE ↓	1496.72	544.68	**535.99**	734.69
Superconductivity	RMSE ↓	9.454	9.302	**9.209**	11.066
Online Shoppers	Accuracy ↑	0.8897	0.9031	**0.9039**	0.8946
	AUC ↑	0.9018	**0.9306**	0.9306	0.9040
Stroke	Accuracy ↑	0.9413	0.9491	0.9501	**0.9511**
	AUC ↑	0.7605	**0.8388**	0.8231	0.2690
HR Attrition	Accuracy ↑	**0.8673**	0.8571	0.8469	0.8639
	AUC ↑	0.7631	0.7552	**0.7886**	0.7512

Classification tasks. For Online Shoppers and Stroke, XGBoost and LightGBM obtain the strongest AUC-ROC scores. xRFM remains reasonably competitive on Online Shoppers, but its AUC is below the boosted-tree models. xRFM's strongest result is on HR Attrition, where it achieves the highest accuracy of 0.8673, suggesting a modest advantage in small-sample classification. MLP obtains high accuracy on Stroke but very low AUC, showing that accuracy alone can be misleading under severe class imbalance.

Table 4 reports training time and per-sample inference time for all models. The timing results show that xRFM can be efficient on smaller datasets, but becomes very expensive on Diamonds. In contrast, LightGBM and XGBoost remain consistently fast.

Table 4: Training time (s) and inference time per sample (μs) for all models. Each cell shows Train / Infer.

Dataset	xRFM	XGBoost	LightGBM	MLP
Diamonds	246.5 s / 69.8 μs	1.4 s / 1.9 μs	0.8 s / 1.9 μs	26.8 s / 0.9 μs
Superconductivity	1.6 s / 1.0 μs	3.4 s / 2.1 μs	1.9 s / 7.6 μs	34.3 s / 0.8 μs
Online Shoppers	0.7 s / 0.8 μs	0.2 s / 1.4 μs	0.3 s / 1.6 μs	2.1 s / 0.8 μs
Stroke	0.3 s / 1.1 μs	0.1 s / 1.1 μs	0.1 s / 2.0 μs	0.4 s / 1.0 μs
HR Attrition	0.1 s / 3.2 μs	0.1 s / 3.0 μs	0.1 s / 7.5 μs	0.4 s / 2.2 μs

3.2 Scalability Analysis

Scalability experiments (detailed in Table 6, Appendix B) reveal a significant efficiency gap between models. As visualised in Figure 1, xRFM exhibits near-quadratic time complexity on the Diamonds dataset, where training time increases exponentially with marginal RMSE gains (reaching 226.44s at $n = 30{,}000$). In contrast, GBDT baselines maintain sub-second training times even at full scale.

However, xRFM scales more effectively on the purely numerical Superconductivity dataset, nearly matching GBDT performance. This divergence confirms that xRFM's AGOP-based reweighting is better suited for continuous high-dimensional inputs than for mixed-type data with sharp categorical thresholds.

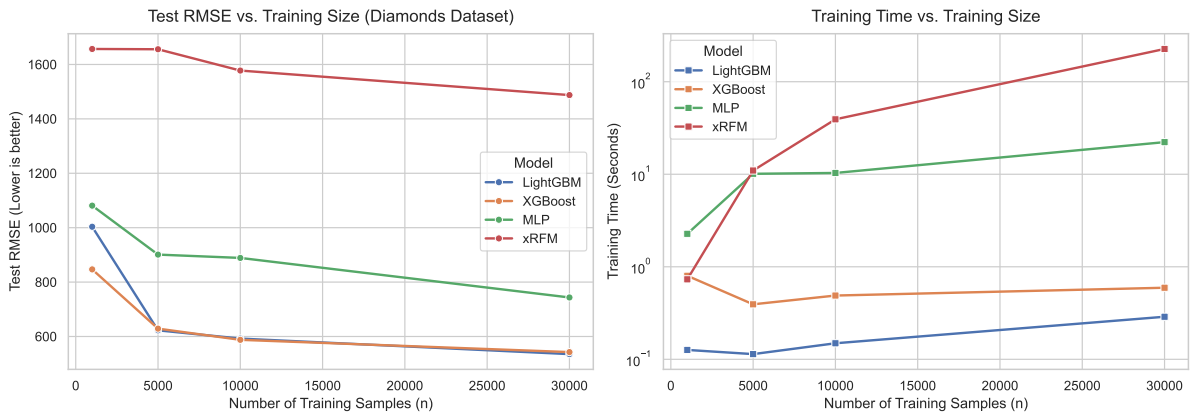

Figure 1: Scalability analysis on the Diamonds dataset, comparing test RMSE (left) and training time (right) across varying training set sizes.

3.3 Interpretability Comparison

AGOP-diagonal rankings (detailed in Table 7, Appendix B) align closely with domain knowledge across all tasks, identifying physical dimensions for Diamonds and health risk factors like glucose and age for Stroke.

To answer whether AGOP agrees with other feature-importance methods, we perform a detailed comparison on the Diamonds dataset (Table 5 and Figure 2).

Table 5: Feature-importance comparison on the Diamonds dataset.

Method	Top ranked features
AGOP diagonal	`y`, `x`, `carat`, `z`, `depth`
PCA loadings	`x`, `y`, `z`, `carat`, `table`
Mutual information	`carat`, `x`, `y`, `z`, `clarity_SI1`
Permutation importance	`carat`, `y`, `x`, `z`, `color_I`

The empirical comparison partly supports the theoretical argument made in Section 2.1. AGOP successfully identifies continuous features (`y`, `x`, `carat`) that the learned xRFM function is highly sensitive to, agreeing with mutual information. However, permutation importance assigns additional weight to categorical quality features (e.g., `color_I`) that AGOP downplays.

This suggests that while AGOP identifies sensible predictive directions, its rankings do not guarantee overall accuracy. On Diamonds, despite identifying relevant size features, xRFM's RMSE remains nearly three times higher than tree-based models. Interpretability and predictive power are decoupled here: AGOP highlights the sensitivity of the fitted model, but the xRFM pipeline itself struggles to capture sharp axis-aligned thresholds and categorical interactions as effectively as boosted trees.

4 Discussion

4.1 xRFM: When Feature Learning Helps and When It Does Not

xRFM's strongest result is HR Attrition ($n = 882$), where its structured feature-learning prior may help in low-data settings. However, on Diamonds it obtains RMSE 1496.72 versus 535.99 for LightGBM — nearly three times worse. Diamonds contains strong axis-aligned categorical effects (cut, colour, clarity) that boosted trees capture efficiently through split-based interactions [Chen

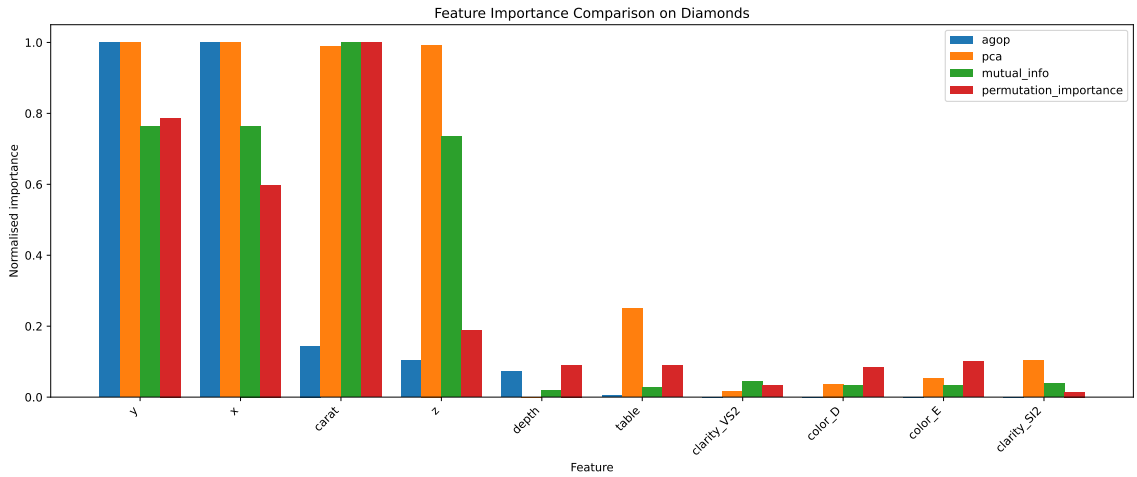

Figure 2: Feature-importance comparison on the Diamonds dataset using AGOP diagonal scores, PCA loadings, mutual information, and permutation importance.

and Guestrin, 2016, Ke et al., 2017], whereas AGOP gradient reweighting is less effective here. The scalability experiment confirms that adding more Diamonds data does not close this gap. In contrast, on the fully numerical Superconductivity dataset xRFM approaches the tree baselines, suggesting it is better suited to continuous high-dimensional inputs. Future work should target categorical handling, more efficient AGOP approximation, and dataset-specific tuning.

4.2 XGBoost and LightGBM: Performance and Efficiency

XGBoost and LightGBM deliver competitive or superior performance across all five datasets while remaining the fastest models to train. On Diamonds, both GBDTs outperform xRFM by a large margin in RMSE (544.68 and 535.99 versus 1496.72), consistent with the strong axis-aligned inductive bias of tree-based splits [Grinsztajn et al., 2022]. On HR Attrition ($n = 882$, $d = 51$), xRFM marginally surpasses XGBoost in accuracy (0.8673 versus 0.8571), suggesting that kernel-style feature learning offers a modest advantage only in a restricted small-data regime. LightGBM is the fastest model at every large Diamonds scale, with near-linear growth in training time (Table **??**).

4.3 MLP: Performance and Scalability Limitations

MLP underperforms both GBDTs on regression (RMSE 743.5 vs. 535.3 on Diamonds at $n =$ 30,000) and shows unstable classification results, most notably AUC 0.269 on Stroke. This reflects sensitivity to class imbalance and the limited alignment between global gradient-based optimisation and the piecewise structure of tabular features [Holzmüller et al., 2024].

5 Conclusion

We replicated xRFM and compared it with XGBoost, LightGBM, and MLP across five tabular datasets under a unified experimental pipeline. Tree-based GBDTs remain the strongest general-purpose baselines: LightGBM and XGBoost lead on accuracy and training speed across nearly all settings. xRFM shows selective promise — competitive on Superconductivity and best accuracy on HR Attrition — but underperforms substantially on Diamonds and scales poorly there, indicating that AGOP feature learning is not universally superior to tree splitting on large mixed-type data. AGOP-diagonal rankings are broadly interpretable and align with domain knowledge, though meaningful rankings do not guarantee predictive strength. Future work should

address categorical variable handling, AGOP approximation efficiency, and hyperparameter tuning for diverse tabular settings.

A Bonus: Residual-Weighted AGOP

(i) Justification. Standard AGOP weights all samples equally ($G_f = \frac{1}{n}\sum_i \nabla f(x_i)\nabla f(x_i)^\top$). Points with large residuals $r_i = y_i - f(x_i)$ indicate regions where the model fails. Upweighting by $\varphi(r_i) = r_i^2$ shifts the AGOP toward directions of high model error, providing a more focused signal for guided tree splits.

(ii) Implementation. The residual-weighted AGOP is defined as

$$\mathrm{AGOP}_{\mathrm{res}}(f) = \frac{\sum_{i=1}^{n} r_i^2 \, \nabla f(x_i)\nabla f(x_i)^\top}{\sum_{i=1}^{n} r_i^2}. \tag{2}$$

Gradients are estimated by forward finite differences ($\varepsilon = 10^{-3}$) on a degree-2 Ridge regression model. The core implementation is two lines:

```
def agop(G, residuals=None):
    w = np.ones(len(G))/len(G) if residuals is None else residuals**2
    w /= w.sum() + 1e-12
    return np.einsum('i,ij,ik->jk', w, G, G)
```

(iii) Disagreement and Performance. On a two-regime dataset, residual-weighted AGOP identifies the signal in the hard-to-fit regime (x_2), whereas standard AGOP focuses on the well-fitted regime (x_0). Although both produce similar test RMSE in a single-level split, residual weighting provides a superior interpretive signal for identifying model failure regions.

B Supplementary Tables

Table 6: Full scalability experiment results: test RMSE and training time (seconds) across varying training set sizes.

Dataset	n_{train}	xRFM		XGBoost		LightGBM		MLP	
		RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)
Diamonds	1,000	1657.0	0.77	846.7	0.80	1003.4	0.13	1081.0	2.27
	5,000	1655.9	11.03	628.9	0.39	622.7	0.11	901.1	10.12
	10,000	1577.6	39.31	587.8	0.49	592.3	0.15	888.9	10.33
	30,000	1487.5	226.44	542.7	0.59	535.3	0.29	743.5	22.23
Supercond.	1,000	14.07	0.08	13.76	0.39	13.91	0.15	15.39	2.52
	5,000	11.30	0.47	10.81	1.00	10.71	0.46	13.05	8.58
	10,000	9.81	1.07	9.51	1.73	9.50	1.10	11.22	28.35
	12,757	9.46	1.56	9.28	1.64	9.21	1.04	–	–

Table 7: Full xRFM AGOP-diagonal feature importance for all evaluated datasets.

Dataset	Top AGOP-ranked features, from 1st to 5th
Diamonds	y, x, carat, z, depth
Superconductivity	number_of_elements, mean_atomic_mass, wtd_mean_atomic_mass, gmean_atomic_mass, wtd_gmean_atomic_mass
Online Shoppers	PageValues, ExitRates, ProductRelated, ProductRelated_Duration, BounceRates
Stroke	avg_glucose_level, age, bmi, work_type_Self-employed, smoking_status_smokes
HR Attrition	OverTime_No, OverTime_Yes, BusinessTravel_Travel_Frequently, TotalWorkingYears, BusinessTravel_Non-Travel

C Supplementary Figures

AGOP Feature Importance

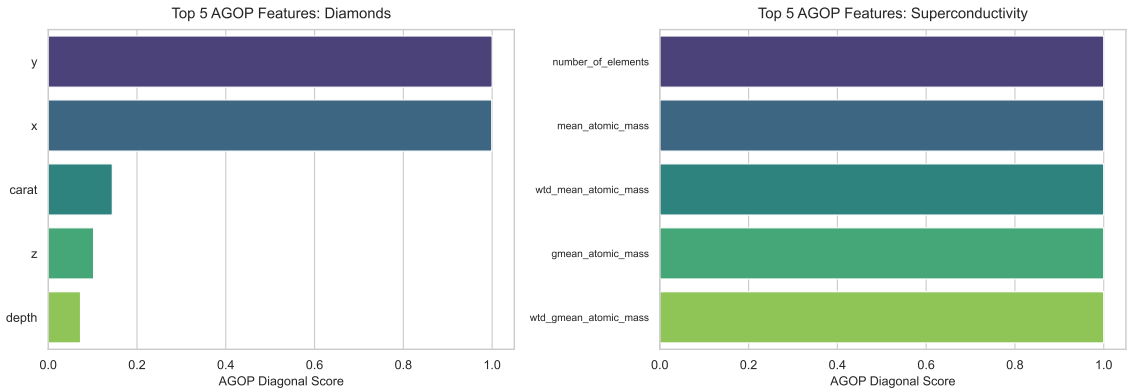

Figure 3: Top-five AGOP diagonal feature importance scores for Diamonds and Superconductivity.

Superconductivity Scalability

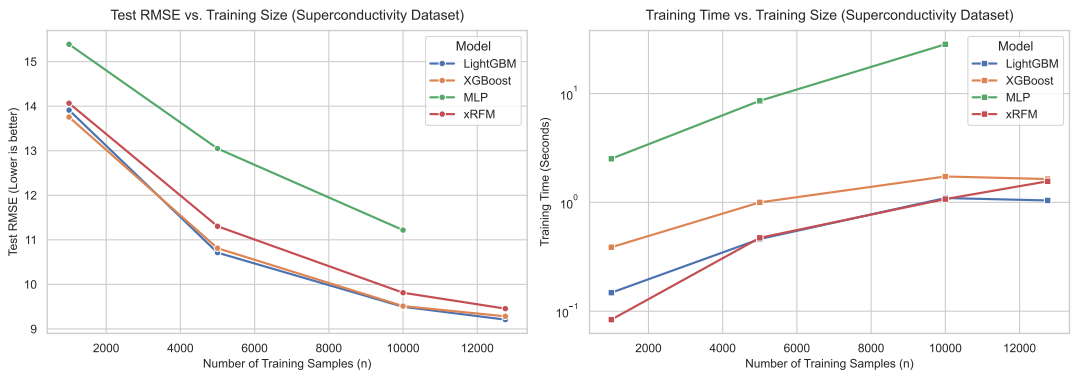

Figure 4: Scalability on Superconductivity: test RMSE and training time vs. training-set size.

Training-Time and Accuracy Trade-off

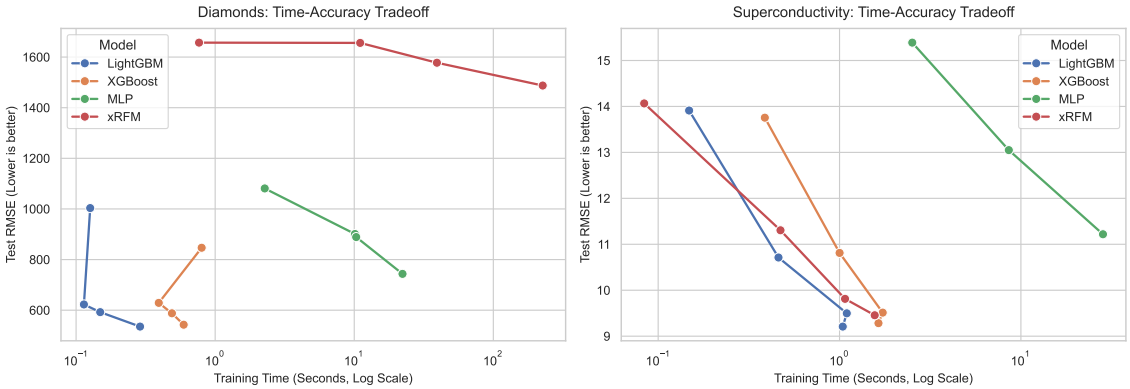

Figure 5: Training-time vs. predictive-error trade-off on Diamonds and Superconductivity.

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